SDM COLLEGE OF ENGINEERING AND TECHNOLOGY
Dharwad-580002
(An autonomous Institution affiliated to Visvesvaraya Technological University, Belagavi – 590018)



Department of Computer Science and Engineering

Academic Year: 2025-26

Internship Report entitled

"System Engineer Internship"

Winter Internship

Course Code 22UCSL801

Submitted by

Name of the Student: Eshwari N Punage

USN: 2SD22CS025

Semester: VIII

SDM COLLEGE OF ENGINEERING AND TECHNOLOGY
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Department of Computer Science and Engineering

CERTIFICATE

This is to certify that Eshwari N Punage bearing USN:2SD22CS025, Student of VIII Semester has satisfactorily completed the Summer Internship entitled "System Engineer Internship" at Department of Computer Science and Engineering, SDM College of Engineering and Technology, Dharwad – 580002 and submitted for the partial fulfilment of the Degree.

Coordinator,
Winter Internship

Head of the Department

External Viva

Name of Examiners **Signature with date**

1. _____
2. _____

CERTIFICATE OF COMPLETION



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CERTIFICATE OF COMPLETION
OF
INTERNSHIP

This is to certify that

Eshwari Punage

Of

SHRI DHARMASTHALA MANJUNATHESHWARA COLLEGE OF ENGINEERING & TECHNOLOGY, DHARWAD, Visvesvaraya Technological University

has completed the internship program at Infosys Limited

from

January, 2026 – May, 2026

Satheesha B N.

Satheesha B Nanjappa

Senior Vice President and Head - Education, Training and Assessment(ETA)

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Infosys Limited for providing me with the opportunity to undergo the Infosys Winternship Program and gain practical exposure to enterprise technologies and professional work culture.

I extend my heartfelt thanks to our faculty guide Prof. Chaitali A Chate and the Department of Computer Science and Engineering, SDMCET, for their valuable guidance, encouragement, and continuous support throughout the internship period.

I would also like to thank the mentors and trainers at Infosys for conducting comprehensive training sessions in Generics and Big Data technologies. Their mentorship helped me strengthen my programming fundamentals, analytical thinking, and understanding of enterprise-level software development practices.

I am grateful to my project teammates for their cooperation, collaboration, and support during the development of the Transaction Reconciliation Engine project. Working as a team helped me improve both my technical and interpersonal skills.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this internship.

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INTERNSHIP SCHEDULE

The internship was conducted for a duration of 18 weeks under the Infosys Winternship Program. The internship was divided into two major phases consisting of structured technical training and project implementation.

Phase 1: Generics Training Program (Initial Training Phase)

Week 1 – Week 4

Introduction to enterprise software development practices.

1. Programming fundamentals and problem-solving techniques.
2. Object-Oriented Programming concepts using Java.Coding standards and debugging techniques.
3. Database fundamentals and SQL concepts.
4. Logical reasoning and analytical skill development.

Phase 2: Big Data Stream Training

Week 5 – Week 12

Introduction to Big Data architecture and ecosystem.

1. Distributed data processing concepts using Apache Spark.
2. Working with HDFS and distributed storage systems.
3. Data warehousing and data lake concepts.
4. Hands-on exposure to MongoDB and NoSQL databases.
5. Advanced SQL concepts including stored procedures, triggers, and cursors.
6. Working with Unix Operating System and enterprise data workflows.
7. Data visualization and dashboard preparation using Power BI.

Phase 3: Project Development – Transaction Reconciliation Engine

Week 13 – Week 18

Week 13:

Project planning and requirement analysis.

1. Database schema design for transaction management and reconciliation.
2. Understanding fintech reconciliation workflows and logging mechanisms.

Week 14:

Development of transaction processing module using Flask and MySQL.

1. Implementation of sender/receiver validation and balance verification.
2. Logging gateway and transaction records into database tables.

Week 15:
Development of reconciliation engine using Apache Spark (PySpark).

1. Implemented mismatch detection, missing transaction detection, and duplicate handling.
2. Added CSV-based reconciliation support.

Week 16:
Developed reconciliation dashboard and results visualization pages.

1. Implemented filters for out-of-order transactions and large transactions.
2. Added ZIP export functionality for transaction logs.

Week 17:
Integration of frontend and backend modules.

1. Performed debugging and testing using sample banking datasets.
2. Improved reconciliation performance and data handling logic.

Week 18:
Final project testing and optimization.

1. Documentation preparation and result validation.
2. Project demonstration and deployment preparation.

INTRODUCTION OF THE COMPANY AND DEPARTMENT

Company Name: Infosys Limited

About the Company

Infosys Limited is one of the leading global technology and consulting companies providing business consulting, information technology, and digital transformation services. The company offers enterprise-scale software solutions, cloud services, data engineering, analytics, and AI-driven technologies to clients across multiple industries.

Infosys is known for its strong focus on innovation, learning, and professional development. Through programs such as the Winternship Internship, students are exposed to enterprise-level workflows, software engineering standards, structured learning methodologies, and real-time project development practices.
The internship program at Infosys bridges the gap between academic learning and industrial implementation by providing hands-on experience in modern technologies and development environments.

ACTIVITIES OR RESPONSIBILITIES

During the internship, I was involved in multiple technical learning activities and project development responsibilities.

Training Activities

1. Attended technical training sessions on Generics and Big Data technologies.
2. Completed coding exercises and technical assignments.
3. Practiced SQL querying and database operations.
4. Worked with distributed data processing concepts using Apache Spark.
5. Understood ETL workflows, data storage systems, and data analytics.
6. Participated in technical discussions and collaborative learning sessions.

Project Responsibilities

1. Designed and developed modules for the Transaction Reconciliation Engine.
2. Developed transaction processing workflows using Flask and MySQL.
3. Implemented reconciliation logic using Apache Spark.
4. Created frontend pages using HTML, CSS, and JavaScript.
5. Worked on database schema design and transaction logging.
6. Implemented CSV export and ZIP download functionalities.
7. Participated in debugging, testing, and optimization of the application.
8. Collaborated with team members during integration and development activities.
9. Managed GitHub-based code collaboration and version control.

MAJOR TASKS / PROJECTS

Transaction Reconciliation Engine (TRE)

Project Overview

The Transaction Reconciliation Engine (TRE) is a fintech-based transaction monitoring and reconciliation system developed using Flask, MySQL, and Apache Spark.

The main objective of the project was to simulate real-world banking reconciliation workflows by comparing external gateway logs with internal transaction records and identifying inconsistencies in transaction processing.

The system was designed to detect:

1. Missing transactions
2. Amount mismatches
3. Duplicate gateway events
4. Large amount transaction records
5. Failed transaction scenarios
6. Rollback and delayed settlement cases

The project supports both CSV-based reconciliation and database-driven reconciliation using Apache Spark for scalable processing.

Technologies Used:

Technology	Purpose
1. Flask	Backend web framework
2. MySQL	Database management
3. Apache Spark (PySpark)	Reconciliation processing
4. HTML/CSS/JavaScript	Frontend development
5. Pandas	CSV handling
6. Git & GitHub	Version control
7. Power BI	Data visualization and analytics

Major Modules Developed

1. Transaction Processing Module

• Developed transaction creation workflow.
• Implemented sender and receiver validation.
• Added transaction status tracking.
• Implemented balance verification logic.
• Added transaction logging.

2. Logging System

• Developed gateway transaction logging mechanism.
• Stored transaction logs into database tables.
• Created export functionality for CSV and ZIP generation.
• Maintained reconciliation job history and status tracking.

3. Reconciliation Engine

• Implemented reconciliation engine using Apache Spark.
• Compared internal transaction logs with gateway logs.
• Implemented mismatch detection and missing transaction detection.
• Developed out-of-order transaction handling logic using timestamps and sequence validation.
• Added large transaction monitoring features.

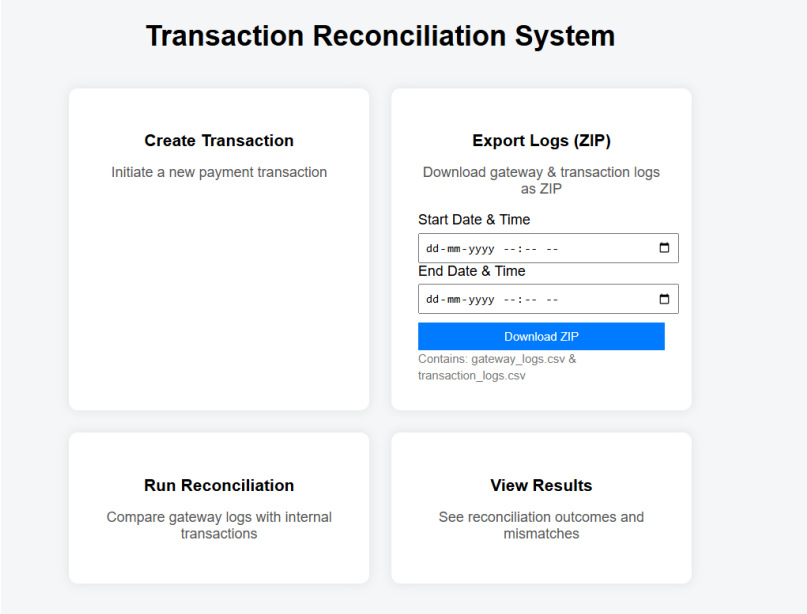
4. Dashboard and Result Visualization

• Developed reconciliation dashboard for displaying reconciliation results.
• Implemented pagination and filtering functionalities.
• Added status-based colour coding for better readability.
• Implemented reconciliation result summary cards.

PROJECT SCREENSHOTS

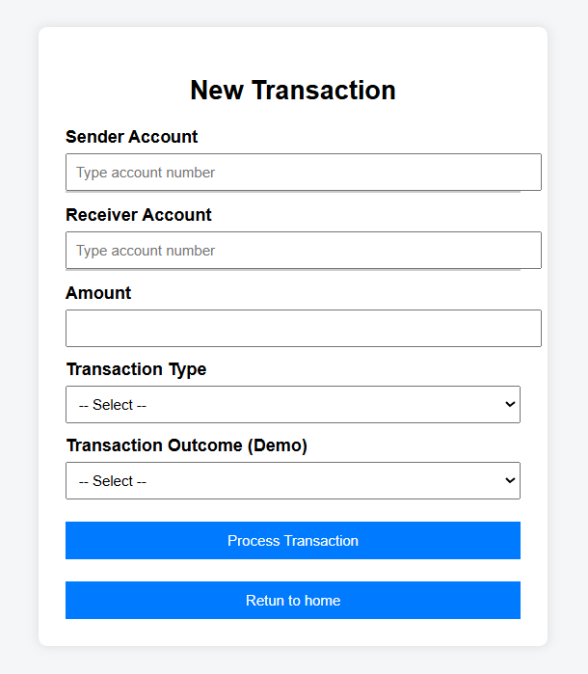
Home Page – Transaction Reconciliation System

This screen acts as the main dashboard of the application where users can create transactions, export logs, run reconciliation processes, and view reconciliation results.

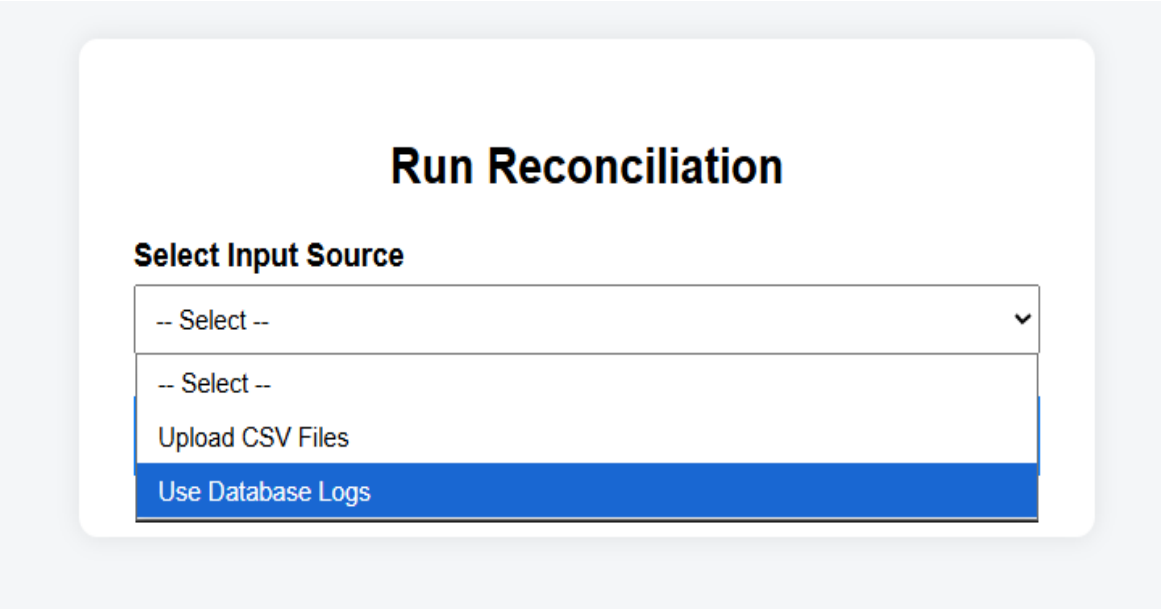


New Transaction Page

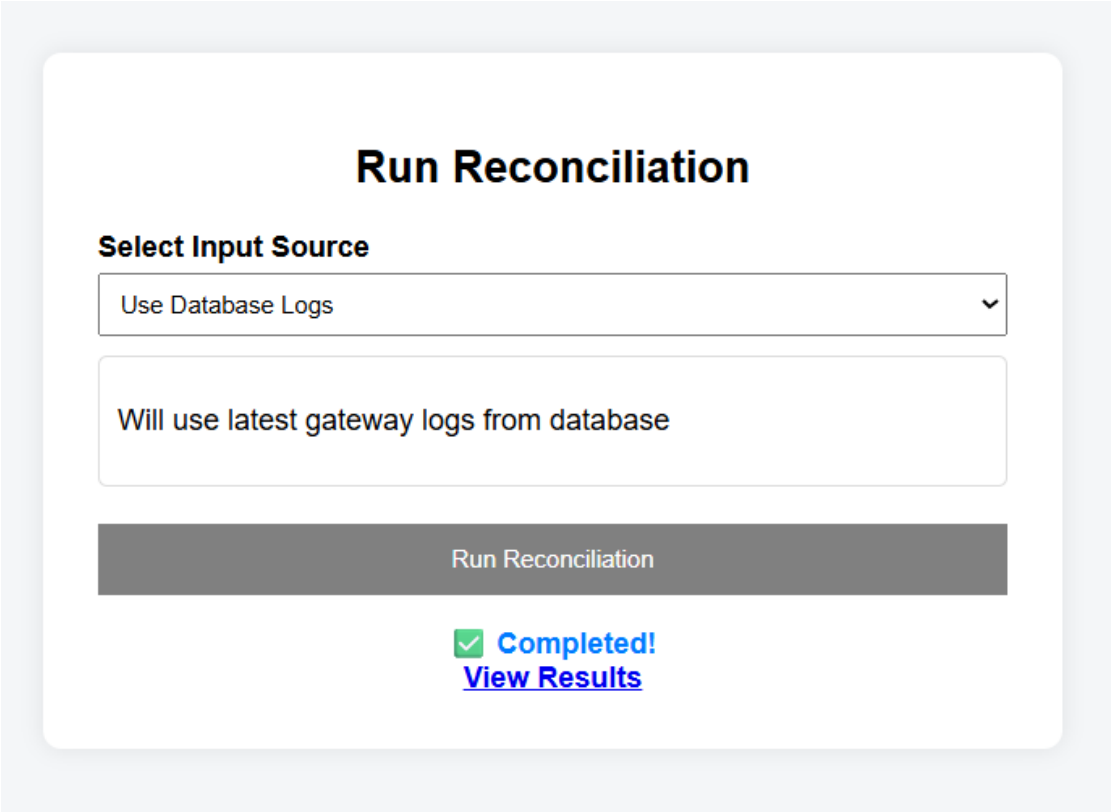
This module allows users to create and process transactions by entering sender account details, receiver account details, amount, transaction type, and transaction outcome.



Reconciliation Input Selection
This screen allows users to select reconciliation input sources either through uploaded CSV files or database logs.

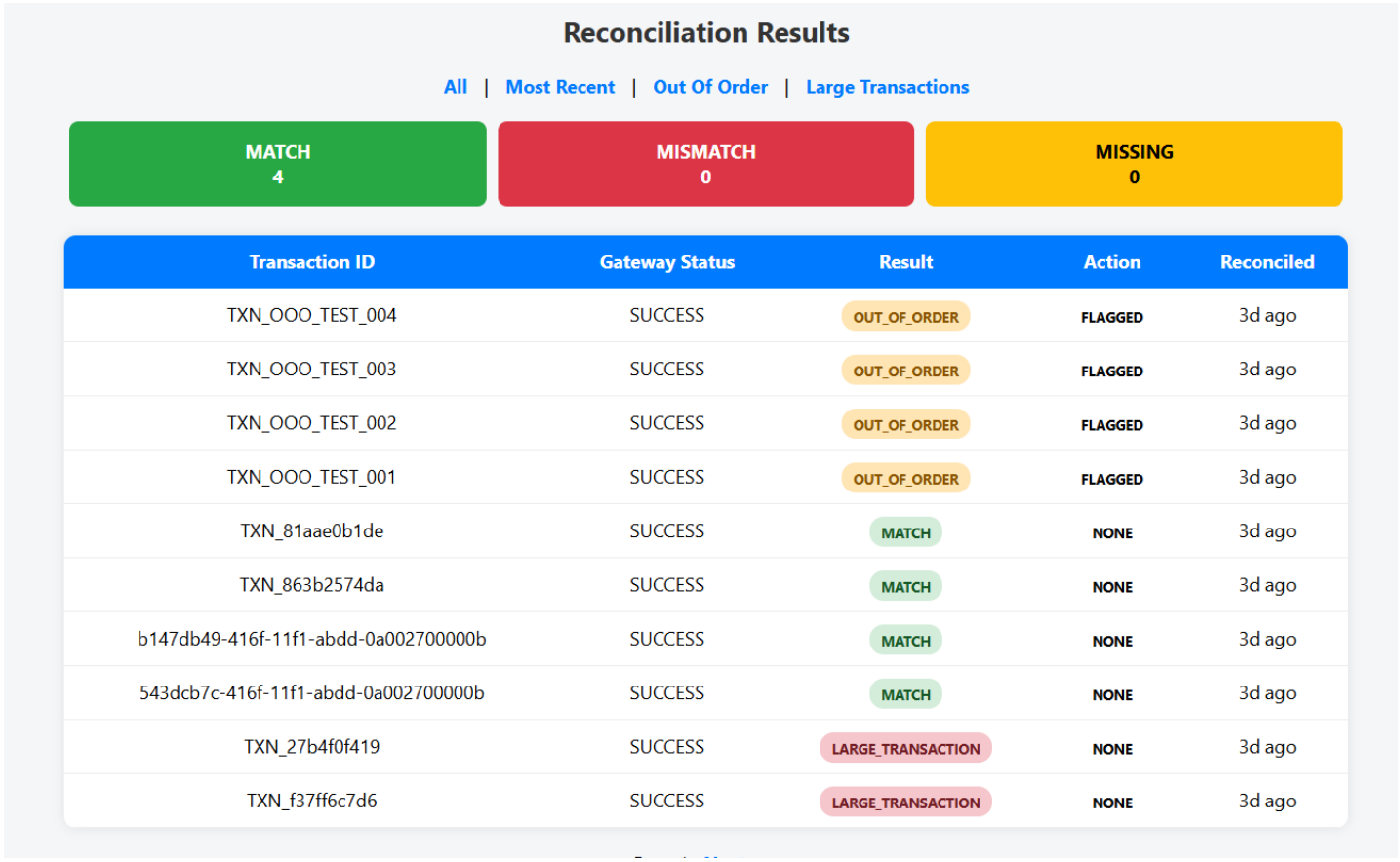


Database Reconciliation Processing
This module performs reconciliation using database logs and Apache Spark processing.



Reconciliation Results Dashboard

The dashboard displays reconciliation outcomes including matched records, out-of-order transactions, and large transaction detections with color-coded indicators.



Reconciliation Results

All | Most Recent | Out Of Order | Large Transactions

MATCH	MISMATCH	MISSING
4	0	0

Transaction ID	Gateway Status	Result	Action	Reconciled
TXN_OOO_TEST_004	SUCCESS	OUT_OF_ORDER	FLAGGED	3d ago
TXN_OOO_TEST_003	SUCCESS	OUT_OF_ORDER	FLAGGED	3d ago
TXN_OOO_TEST_002	SUCCESS	OUT_OF_ORDER	FLAGGED	3d ago
TXN_OOO_TEST_001	SUCCESS	OUT_OF_ORDER	FLAGGED	3d ago
TXN_81aae0b1de	SUCCESS	MATCH	NONE	3d ago
TXN_863b2574da	SUCCESS	MATCH	NONE	3d ago
b147db49-416f-11f1-abdd-0a002700000b	SUCCESS	MATCH	NONE	3d ago
543dcb7c-416f-11f1-abdd-0a002700000b	SUCCESS	MATCH	NONE	3d ago
TXN_27b4f0f419	SUCCESS	LARGE_TRANSACTION	NONE	3d ago
TXN_f37ff6c7d6	SUCCESS	LARGE_TRANSACTION	NONE	3d ago

KEY LEARNINGS

Throughout the internship, I gained strong technical knowledge along with professional development skills.

Technical Learnings

1. Understanding enterprise software development workflows.
2. Hands-on experience with Apache Spark and distributed data processing.
3. Backend development using Flask and Python.
4. Database design and SQL query optimization.
5. Real-time transaction processing and reconciliation concepts.
6. Understanding fintech transaction workflows and audit logging.
7. CSV processing and data transformation techniques.
8. Frontend development using HTML, CSS, and JavaScript.
9. Version control and collaborative development using Git and GitHub.
10. Exposure to Power BI and data visualization concepts.

Professional Learnings

1. Team collaboration and communication.
2. Problem-solving and debugging techniques.
3. Documentation and project planning.
4. Time management and task prioritization.
5. Understanding enterprise coding standards and workflows.
6. Working in an iterative development environment.

Outcomes

1. Successfully completed Generics and Big Data training programs.
2. Developed a functional Transaction Reconciliation Engine application.
3. Implemented Apache Spark based reconciliation workflows.
4. Built reconciliation dashboards and transaction management modules.
5. Improved understanding of enterprise software architecture and distributed systems.

CHALLENGES FACED

Technical Challenges:

Real-Time Reconciliation Logic

Developing the reconciliation engine required handling multiple transaction scenarios such as mismatches, missing records, out-of-order transactions, and delayed gateway events.

Apache Spark Integration

Integrating Apache Spark with Flask and MySQL required proper configuration of Spark sessions, JDBC connectors, and distributed processing workflows.

Database Design

Designing normalized database tables for transaction logs, gateway logs, reconciliation jobs, and audit trails required careful planning.

CSV and Database Synchronization

Ensuring consistency between uploaded CSV files and database-driven reconciliation workflows was challenging during testing and validation.

UI and Dashboard Design

Creating a clean and readable reconciliation dashboard with filters, pagination, and status indicators required multiple iterations and UI refinements.

Collaboration Challenges:

Team Coordination

Coordinating development tasks among team members and integrating modules into a unified application required proper communication and version control practices.

Merge Conflict Resolution

Working with multiple branches in GitHub occasionally caused merge conflicts which needed careful debugging and resolution.

Debugging Complex Scenarios

Testing reconciliation workflows using different transaction cases and failure scenarios required extensive debugging and validation.

CONCLUSION

The Infosys Winternship Program has been a highly valuable learning experience that provided me with both technical knowledge and practical industry exposure. The internship helped me strengthen my programming fundamentals, understand enterprise development practices, and gain hands-on experience in Big Data technologies and distributed data processing.

Working on the Transaction Reconciliation Engine project allowed me to apply theoretical concepts to a real-world fintech-oriented application. Through this project, I gained practical exposure to backend development, database management, Apache Spark processing, transaction reconciliation workflows, and frontend integration.

The internship also enhanced my teamwork, communication, debugging, and problem-solving abilities. Overall, this experience has significantly improved my confidence in software development and prepared me for future opportunities in enterprise application development, Big Data engineering, and fintech systems.

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